Topic 3: Conceptualising Interaction (part 1)

SCSM2113 Human-Computer Interaction Fundamentals

Faculty of Computing
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01 CONCEPTUAL MODELS

02 INTERFACE METAPHORS

03 INTERACTION TYPES

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CONCEPTUAL MODEL

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Conceptual Model

- A conceptual model is:

  "...a high-level description of how a system is organised and operates" (Johnson and Henderson, 2002, p26)

- It enables:

  "...designers to straighten out their thinking before they start laying out their widgets" (Johnson and Henderson, 2002, p28)

- Provides a working strategy and framework of general concepts and their interrelations

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Components

- Metaphors and analogies
  - Understand what a product is for and how to use it for an activity
- Concepts that people are exposed to through the product
  - Task–Domain objects, their attributes, and operations (for example, saving, revisiting, organising)
- Relationship and mappings between these concepts
  - E.g. whether one object contains one another, and, whether one can revisit through looking at a list of visited sites

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UTM

5

...what concepts are needed to understand how to interact with this interface..?



Mappings?
Insert screwdriver
Turning screwdriver

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UTM

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First Steps in Formulating a Conceptual Model

- What will people be doing when carrying out their tasks?
- How will the proposed model support these?
- What kind of interface metaphor, if any, will be appropriate?
- What kinds of interaction modes and styles to use?
- Keep in mind when making design decisions how the user will understand the underlying conceptual model

Conceptual Models

- Many kinds and ways of classifying them

- The best conceptual models are often those that appear:
  - Obvious and simple
  - The operations they support are intuitive to use

INTERFACE METAPHORS

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Interface Metaphors

- Interface designed to be similar to a physical entity but also has own properties
  - e.g, desktop metaphor, web portals, search engine
- Can be based on activity, object, or a combination of both
- Exploit user's familiar knowledge, helping them to understand 'the unfamiliar'
- Conjures up the essence of the unfamiliar activity, enabling users to leverage this to understand more aspects of the unfamiliar functionality

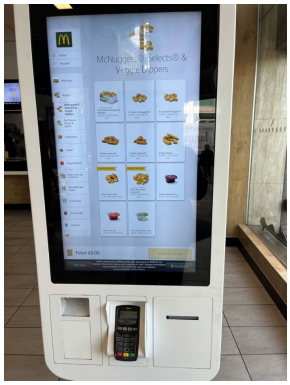
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Examples of Interface Metaphors

- Conceptualising what users are doing
  - e.g, surfing the Web
- A conceptual model instantiated at the interface
  - e.g., the desktop metaphor
- Visualising an operation
  - e.g., an icon of a shopping cart into which the user places items

The Card Metaphor

- The card is a very popular metaphor. Why?
  - It has familiar form factor
  - It can easily be flicked through, sorted, and themed
  - Cards have strong associations, providing an intuitive way of organising content that is "card sized"
  - Structures content into meaningful chunks



A fast food display

Benefits of Interface Metaphors

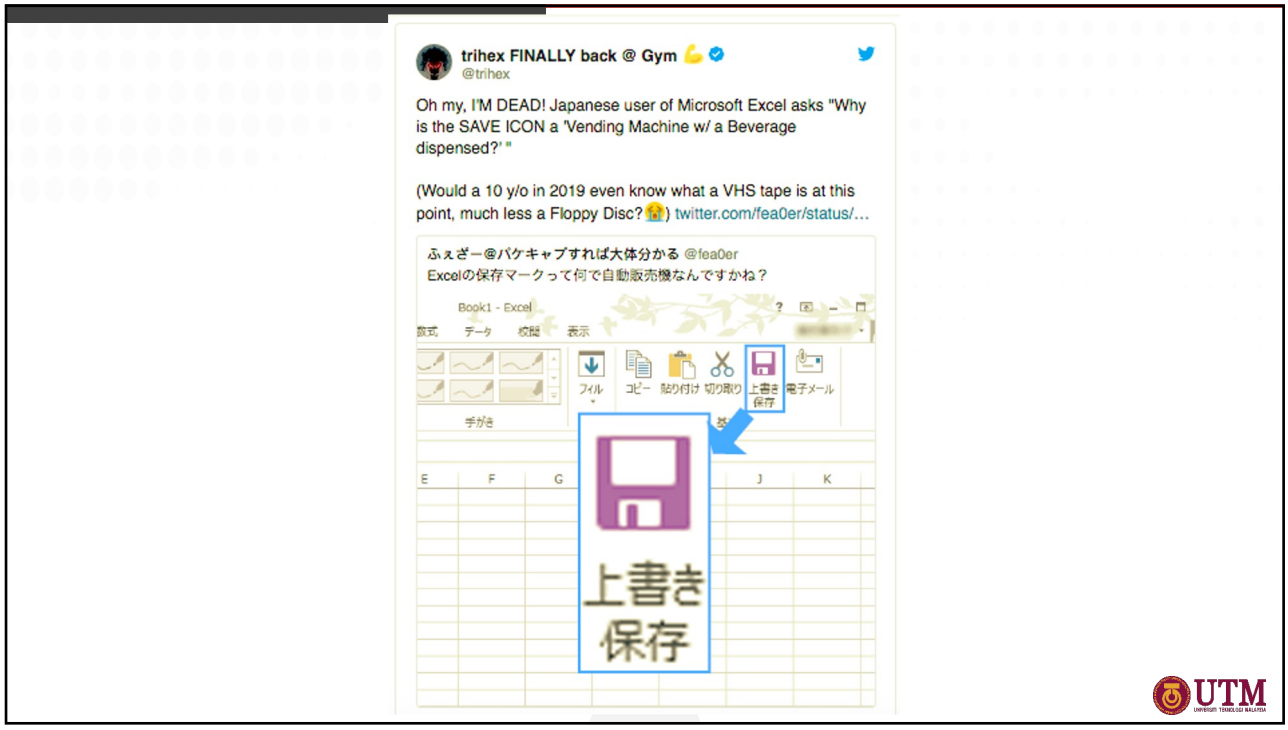
- Makes learning new systems easier
- Familiar artefact or action helps people understand the underlying conceptual model
- Can be innovative and enable apps to be made accessible to a greater diversity of people

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Problems with Interface Metaphors

- Break conventional and cultural rules
  - e.g, recycle bin placed on desktop
- Can constrain designers in the way that they conceptualize a problem space
- Can conflict with design principles
- Designers can inadvertently use bad existing designs and transfer the bad parts over
- Limits designers' imagination in coming up with new conceptual models
- Often don't scale very well, especially when trying to describe how complex systems work (Cooper, 2020)

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INTERACTION TYPES

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Interaction Types

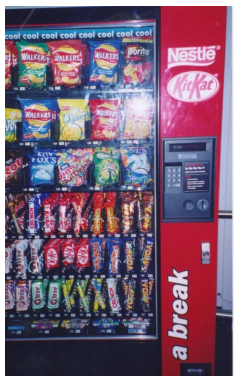
1. Instructing
   - Issuing commands and selecting options
2. Conversing
   - Interacting with a system as if having a conversation
3. Manipulating
   - Interacting with objects in a virtual or physical space by manipulating them
4. Exploring
   - Moving through a virtual environment or a physical space
5. Responding
   - The system initiates the interaction and the person chooses whether to respond

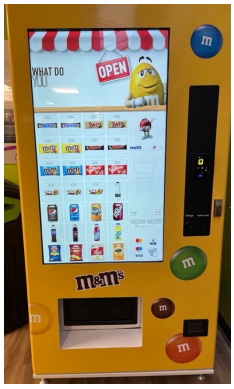
1. Instructing

- Where someone instruct a system and tells it what to do
  - e.g.Tell the time, print a file, or save a file
- Very common conceptual model underlying a diversity of devices and systems
  - e.g. Word processors, VCRs, vending machines
- The main benefit is that instructing supports quick and efficient interaction
  - Good for repetitive kinds of actions performed on multiple objects

Which is easiest and why?





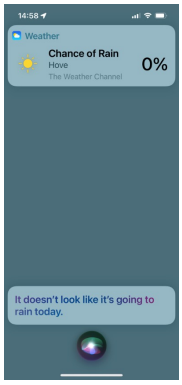


2. Conversing

- Underlying model of having a conversation with another human
- Ranges from simple voice recognition menu-driven systems to more complex 'natural language' dialogs
- Examples include timetables, search engines, advice-giving systems, help systems, virtual agents, chatbots, toys, and pet robots designed to converse with people

Pros and Cons of Conversational Model

- Allows people to interact with a system in a way that is familiar to them
  - can make them feel comfortable, at ease, and less scared
- Misunderstandings can arise when the system does not know how to parse what someone says
  - e.g, voice assistants can misunderstand what children say



Siri's response when asked if a person needs an umbrella today

Pros and Cons of Conversational Model



**"If you'd like to press 1, press 3.
If you'd like to press 3, press 8.
If you'd like to press 8, press 5..."**

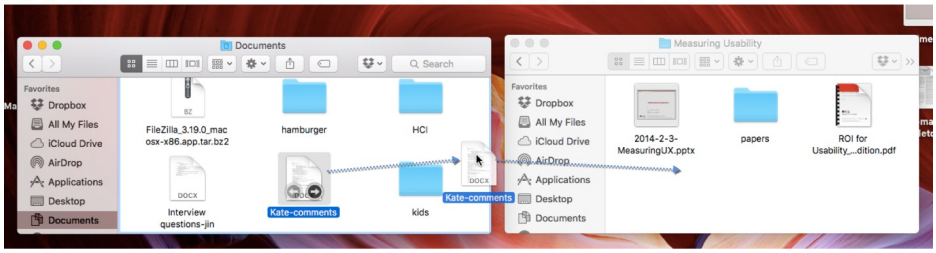
3. Manipulating

- Involves dragging, selecting, opening, closing and zooming actions on virtual objects
- Exploit's someone's knowledge of how they move and manipulate in the physical world
- Can involve actions using physical controllers or air gestures to control the movements of an on-screen avatar
- Tagged physical objects (for instance, balls) that are manipulated in a physical world result in physical/digital events (such as animation)

Direct Manipulation

- Ben Shneiderman (1983) coined the term

- Three core properties:
  - Continuous representation of objects and actions of interest
  - Physical actions and button pressing instead of issuing commands with complex syntax
  - Rapid reversible actions with immediate feedback on object of interest



Moving a file on MacOS using direct manipulation involves dragging that file from the source folder and moving it into the destination folder.

https://www.nngroup.com/articles/direct-manipulation/

Benefits of Direct Manipulation

- Novices can learn the basic functionality quickly
- Experienced users can carry out a wide range of tasks efficiently and rapidly
- Intermittent users can retain operational concepts over time
- Error messages rarely needed
- Users can immediately see if their actions are furthering their goals, and if not, do something else
- Users experience less anxiety
- Users gain confidence and mastery and feel in control

Disadvantages

- Some people take the metaphor of direct manipulation too literally
- Not all tasks can be described by objects, and not all actions can be done directly
- Some tasks are better achieved through delegating, for example, spell checking
- Can become screen space 'gobblers'
- Moving a cursor using a mouse or touchpad can be slower than pressing function keys to do the same actions

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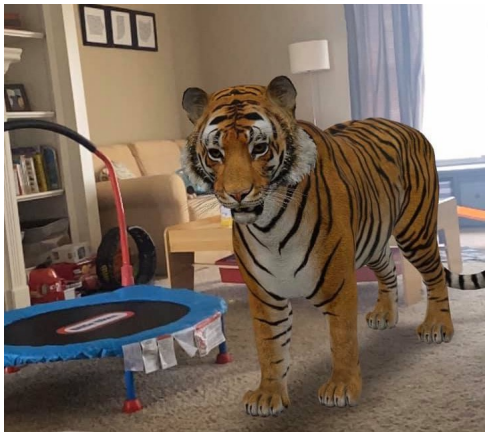
4. Exploring

- Involves moving through virtual or physical environments
  - People can explore aspects of a 3D environment
  - Physical environments can also be embedded with sensors that when detect the presence of someone will trigger digital or physical events to happen

- Many examples of virtual environments, including cities, parks, buildings, rooms, and datasets
  - enable people to fly over them and zoom in and out of different parts

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Seeing things larger than life and in situ in AR



A virtual lion appearing in someone's living room created with Google 3D object

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Exploring Data in VR



Someone using Immersive data analytics allowing them to explore Tableau data in 3D

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5. Responding

- This type suggested by Christopher Lueg et al. (2018)
- System takes the initiative to alert user to something that it "thinks" is of interest
- System does this by:
  - detecting the location and-or presence of someone in a vicinity and notifying them on their phone or smartwatch
  - what it has learned from their repeated behaviors
- Examples:
  - alerts the user of a nearby coffee bar where some friends are meeting
  - A wearer's fitness tracker notifies them of a milestone reached
- Automatic system response without any requests made by the user

Potentials Cons of System-initiated Notifications

- Can get tiresome or frustrating if too many notifications or the system gets it wrong

- What does it do when it gets something wrong?
  - Does it apologise?
  - Does it allow the user to correct the advice or information?

Choosing an Interaction Type

- Direct manipulation is good for 'doing' types of tasks, for example, designing, drawing, flying, driving, or sizing windows
- Issuing instructions is good for repetitive tasks, for example, spell-checking and file management
- Having a conversation is good for finding out information, requesting music/TV or learning
- Hybrid conceptual models are good for supporting multiple ways of carrying out the same actions

Difference between Interaction Types and Interface Styles

- Interaction type:
  - A description of what a person is doing when interacting with a system, for example, instructing, talking, browsing, or responding
- Interface style:
  - The kind of interface used to support the interaction, for instance, command, menu-based, gesture, or voice

Many Kinds of Interface Styles Available (more next topic)

- Command
- Speech
- Data-entry
- Form fill-in
- Query
- Graphical
- Web
- Pen
- Augmented reality
- Gesture

Which Interaction Type to Choose?

- Need to determine requirements and user needs
- Take budget and other constraints into account
- Also will depend on suitability of technology for activity being supported

Summary

- A conceptual model is a high-level description of a product in terms of:
  - What users can do with it and the concepts they need to understand how to interact with it
- Interaction types provide a way of thinking about how to support user's activities